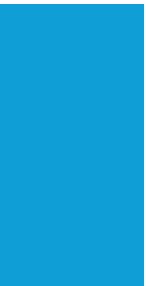


Data Migration Quality Analysis

Presented by Jeffrey Benson





01	Process
02	Findings
03	Interactive Samples
04	Next Steps

Agenda

Data Pipeline Process (ELT)



Sources	Pre-Migration Source	Delta	Post-Migration New
Beneficiary Summary Records	343,644	0	343,644
Carrier Claim Records	4,741,335	+4,777	4,746,112

Capability of Migration Process

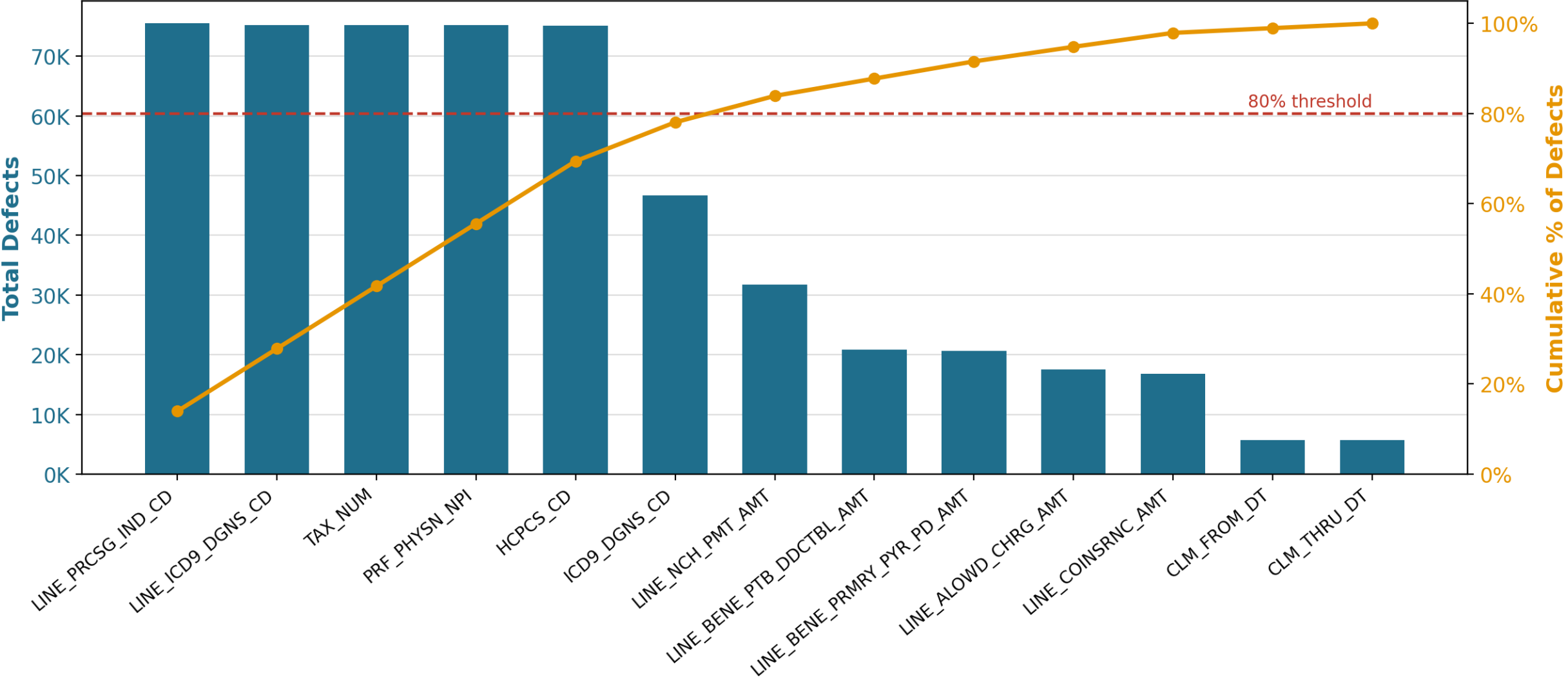
Data Set	Total Rows	Defective Data Elements	Total Data Elements	DPMO*	Process Yield	Sigma
Carrier Claims	4,746.1K	530.4K	484,150.9K	1095.5	99.89045%	~4.56
Beneficiary Summary	343.6K	8.2K	10,657.9K	768.7	99.92313%	~4.67

“Six Sigma” represents a process capability with a 99.99966% yield, or 3.4 defects per million opportunities.

Data Migration Process Capability:

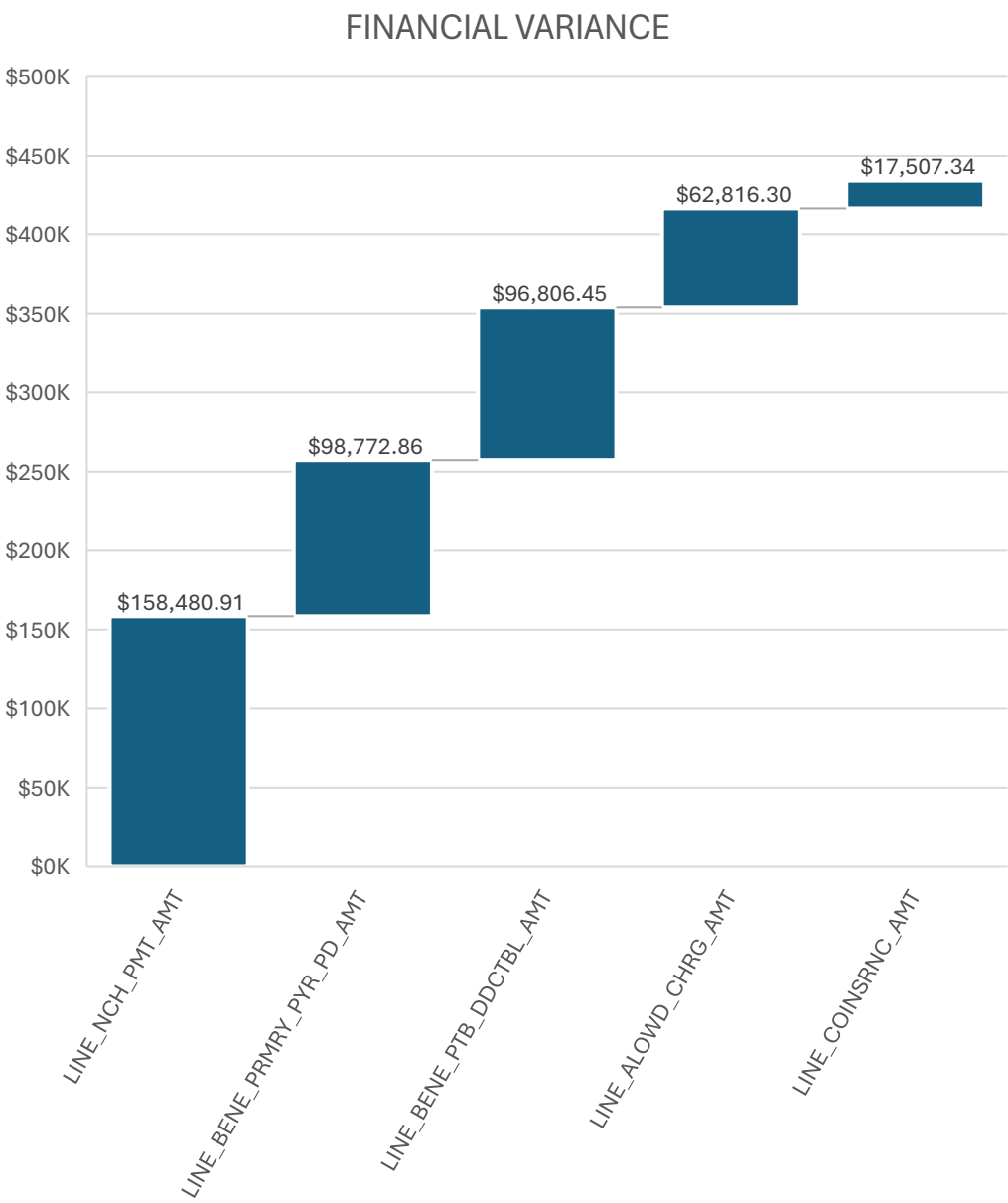
- Carrier Claims: 99.89045% yield, or 1,095.5 defects per million opportunities*.
- Beneficiary Summary: 99.92313% yield, or 768.7 defects per million opportunities*.

Carrier Claims — Defect Pareto by Field



Financial Impact of Claims Data Defects

DATA SET	FINANCIAL VARIANCE	RUNNING PCT OF TOTAL
LINE_NCH_PMT_AMT	\$ 158,480.91	36.48%
LINE_BENE_PRMRY_PYR_PD_AMT	\$ 98,772.86	59.22%
LINE_BENE_PTB_DDCTBL_AMT	\$ 96,806.45	81.51%
LINE_ALOWD_CHRG_AMT	\$ 62,816.30	95.97%
LINE_COINSRNC_AMT	\$ 17,507.34	100.00%
TOTAL	\$ 434,383.86	100.00%



SOURCE	DATA SET	TOTAL DEFECTS	RUNNING PCT	DPMO	SIGMA LEVEL
Beneficiary Summary	BENE_BIRTH_DT	496	6.05%	1,443	4.498
Beneficiary Summary	BENE_HI_CVRAGE_TOT_MONS	337	10.17%	980	4.613
Beneficiary Summary	BENE_SMI_CVRAGE_TOT_MONS	322	14.10%	937	4.626
Beneficiary Summary	BENE_COUNTY_CD	318	17.98%	925	4.629
Beneficiary Summary	BENE_DEATH_DT	318	21.86%	925	4.629
Beneficiary Summary	BENE_ESRD_IND	318	25.74%	925	4.629
Beneficiary Summary	BENE_RACE_CD	318	29.62%	925	4.629
Beneficiary Summary	BENE_SEX_IDENT_CD	318	33.50%	925	4.629
Beneficiary Summary	SP_ALZHDMTA	318	37.39%	925	4.629
Beneficiary Summary	SP_CHF	318	41.27%	925	4.629
Beneficiary Summary	SP_CHRNKIDN	318	45.15%	925	4.629
Beneficiary Summary	SP_CNCR	318	49.03%	925	4.629
Beneficiary Summary	SP_COPD	318	52.91%	925	4.629
Beneficiary Summary	SP_DEPRESSN	318	56.79%	925	4.629
Beneficiary Summary	SP_DIABETES	318	60.67%	925	4.629
Beneficiary Summary	SP_ISCHMCHT	318	64.56%	925	4.629
Beneficiary Summary	SP_OSTEOPRS	318	68.44%	925	4.629
Beneficiary Summary	SP_RA_OA	318	72.32%	925	4.629
Beneficiary Summary	SP_STATE_CODE	318	76.20%	925	4.629
Beneficiary Summary	SP_STRKETIA	318	80.08%	925	4.629
Beneficiary Summary	PLAN_CVRG_MOS_NUM	289	83.61%	841	4.657
Beneficiary Summary	BENRES_CAR	272	86.93%	791	4.674
Beneficiary Summary	MEDREIMB_CAR	260	90.10%	756	4.687
Beneficiary Summary	MEDREIMB_OP	215	92.73%	625	4.740
Beneficiary Summary	BENRES_OP	195	95.11%	567	4.766
Beneficiary Summary	BENE_HMO_CVRAGE_TOT_MONS	124	96.62%	361	4.886
Beneficiary Summary	PPPYMT_CAR	78	97.57%	227	5.002
Beneficiary Summary	MEDREIMB_IP	68	98.40%	198	5.035
Beneficiary Summary	BENRES_IP	62	99.16%	180	5.057
Beneficiary Summary	PPPYMT_IP	36	99.60%	105	5.182
Beneficiary Summary	PPPYMT_OP	33	100.00%	96	5.201

Pareto Analysis of Beneficiary Data Defects

1,322 Beneficiary Records do not match between the Pre- & Post-Migration tables.

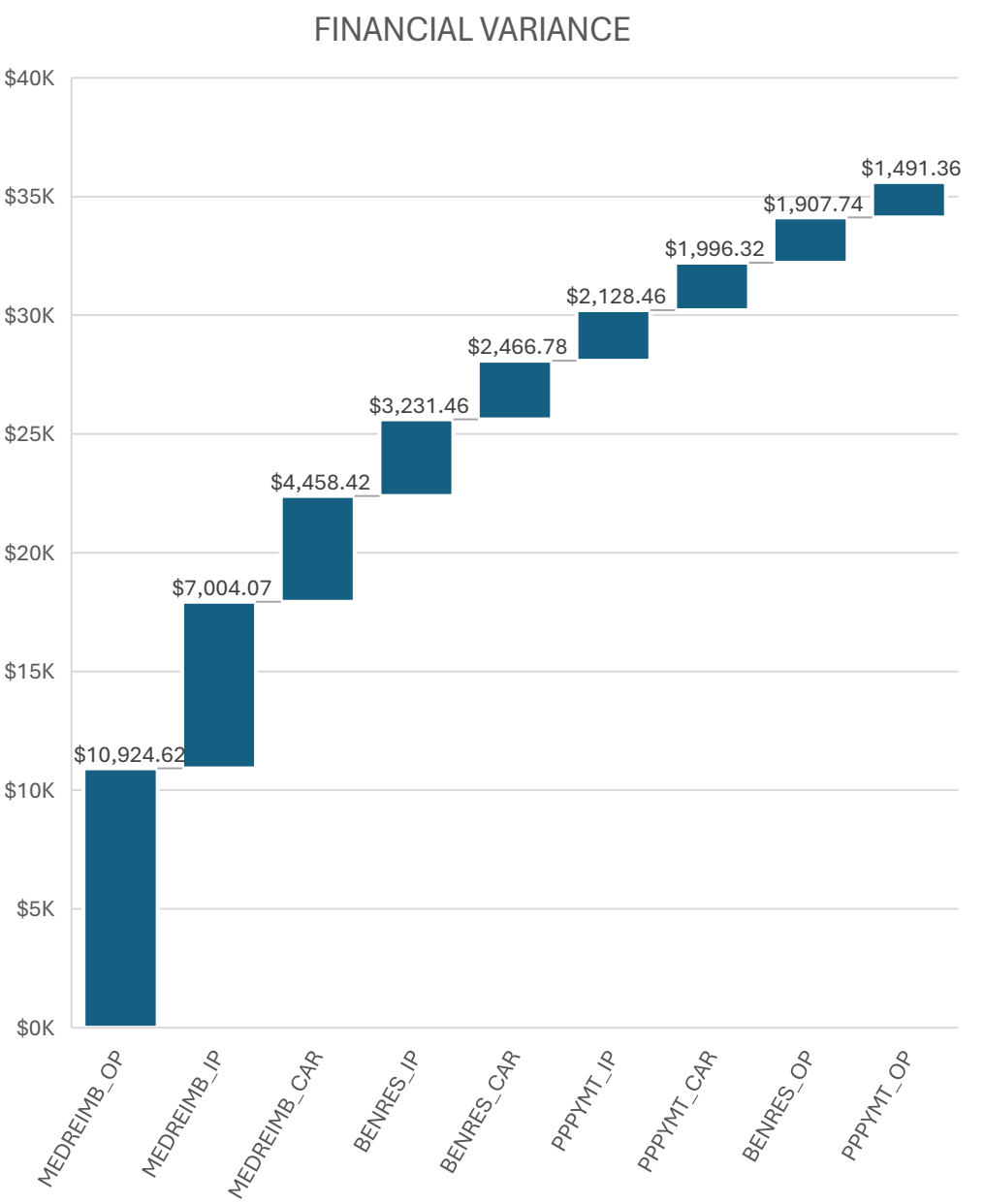
One or more data fields are out of sync.

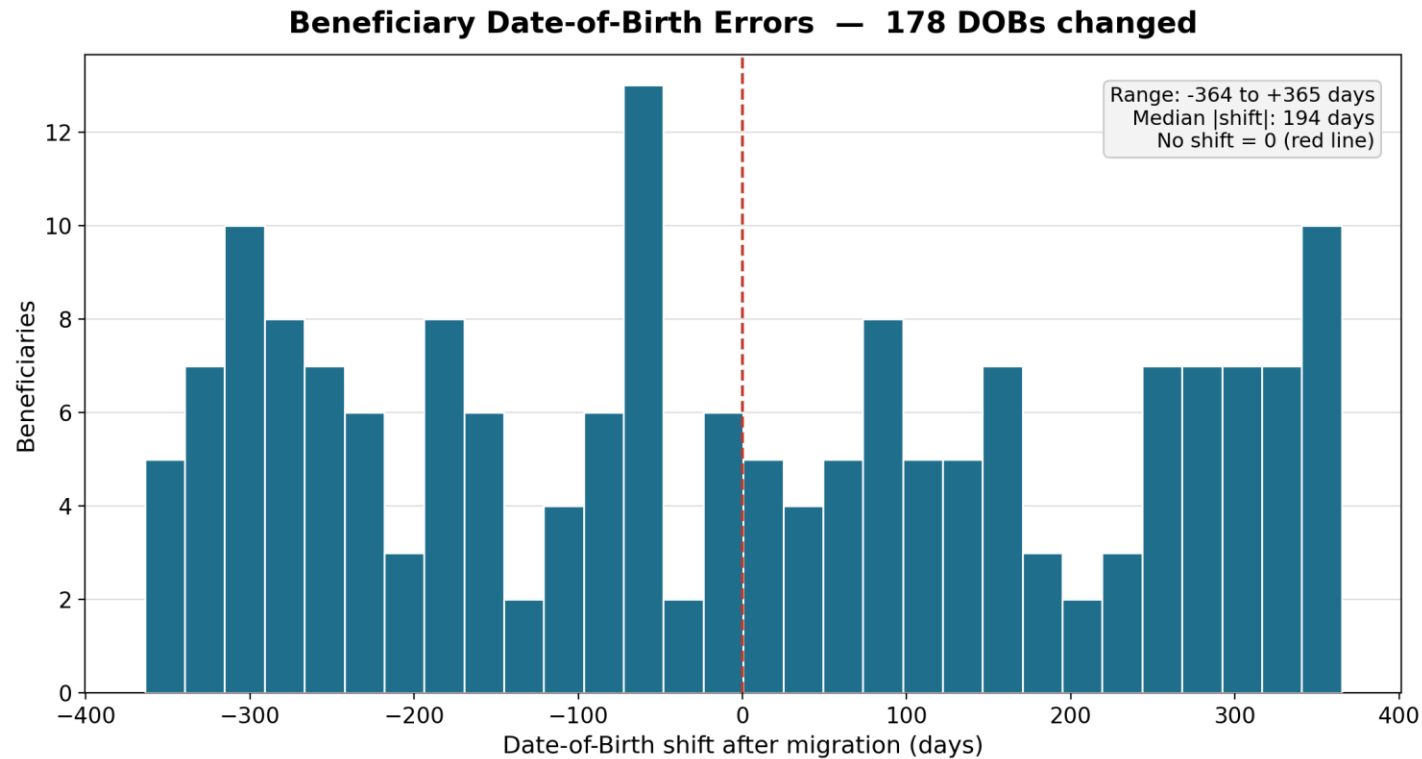
Defects are distributed almost uniformly across ~25 fields (~318 each), indicating a **systemic migration fault affecting most fields equally**—not a few broken columns.

This points root-cause analysis at the migration process itself rather than specific fields.

Financial Impact of Beneficiary Data Defects

DATA SET	FINANCIAL VARIANCE	RUNNING PCT OF TOTAL
MEDREIMB_OP	\$ 10,924.62	30.68%
MEDREIMB_IP	\$ 7,004.07	50.35%
MEDREIMB_CAR	\$ 4,458.42	62.87%
BENRES_IP	\$ 3,231.46	71.94%
BENRES_CAR	\$ 2,466.78	78.87%
PPPYMT_IP	\$ 2,128.46	84.85%
PPPYMT_CAR	\$ 1,996.32	90.45%
BENRES_OP	\$ 1,907.74	95.81%
PPPYMT_OP	\$ 1,491.36	100.00%
TOTAL	\$ 35,609.23	100.00%

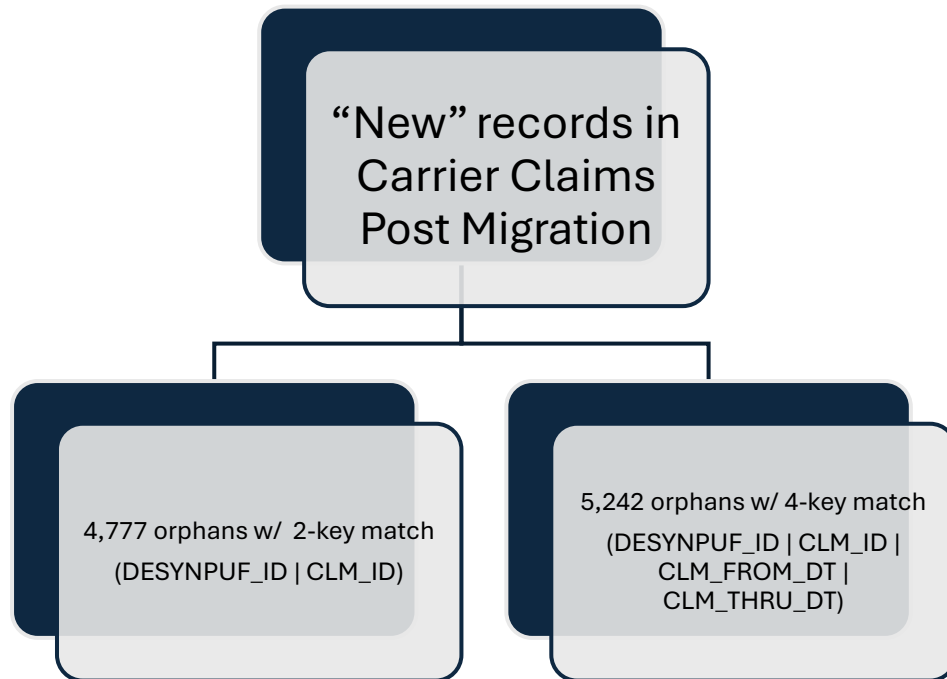




Beneficiary “Date of Birth” Errors

178 Beneficiary
“Dates of Birth”
changed after the
Migration

Carrier Claims Orphan Analysis



Orphans Increased w/ 4-key match due to defects in CLM_FROM_DT and CLM_THRU_DT Post-Migration

Beneficiary ID Errors



159 Missing
in New File



159 Missing
in Original file

Need to Investigate: If same persons, why did the IDs change post migration?

Interactive Dashboard – Financial Pareto



Data Migration Quality Dashboard

CMS DE-SynPUF · Legacy → New System · Six Sigma Analysis · DuckDB + Streamlit

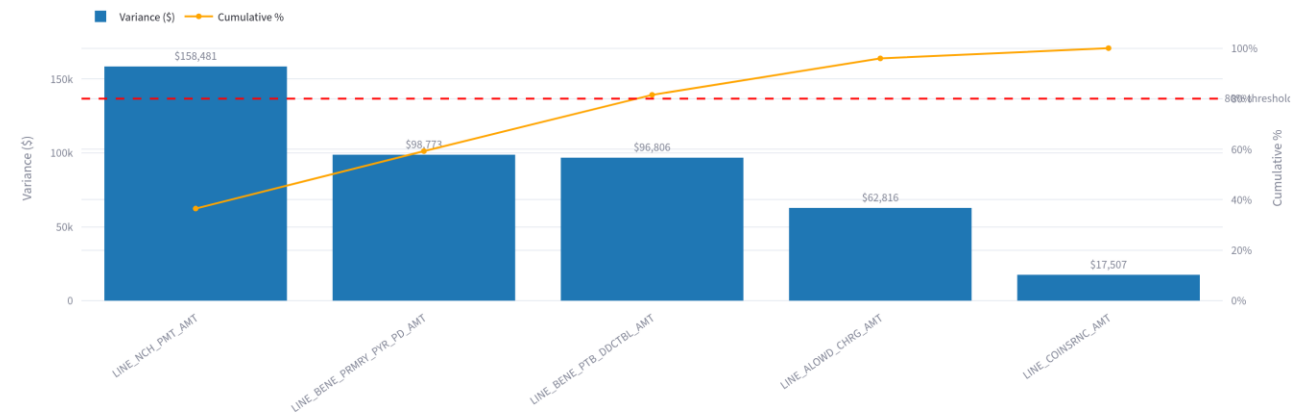
[Migration Scorecard](#) [Six Sigma by Field](#) [Financial Impact](#) [Discrepancy Drill-Down](#) [Row Count Validation](#) [Defect Heatmap](#) [Chronic Condition Errors](#) [Year-over-Year Trend](#) [Go / No-Go](#)

Financial Variance — Pareto Analysis

Dataset

☒ Carrier Claims ☐ Beneficiary Summary

Financial Variance by Field — Carrier Claims (Total: \$434,383.86)



Total Financial Variance

\$434,383.86

Fields driving 80% of variance

2 of 5

Interactive Dashboard – Scorecard



Data Migration Quality Dashboard

CMS DE-SynPUF · Legacy → New System · Six Sigma Analysis · DuckDB + Streamlit

[Migration Scorecard](#) [Six Sigma by Field](#) [Financial Impact](#) [Discrepancy Drill-Down](#) [Row Count Validation](#) [Defect Heatmap](#) [Chronic Condition Errors](#) [Year-over-Year Trend](#) [Go / No-Go](#)

Overall Migration Quality



Carrier Claims

Sigma Level

4.56

Process Yield

99.8904%

DPMO

1,095

Total units: 4,746,577 · Defects: 530,376.0



Beneficiary Summary

Sigma Level

4.67

Process Yield

99.9231%

DPMO

769

Total units: 343,803 · Defects: 8,193.0

Sigma Level Reference

sigma_level	dpmo	yield_pct	status
6σ	3.4	99.9997%	🏆 World-class
5σ	233	99.977%	🟢 Excellent
4.5σ	1,350	99.865%	🟢 Strong
4σ	6,210	99.379%	🟡 Acceptable
3σ	66,807	93.319%	🔴 Needs work

Interactive Dashboard – Go/No-Go



Data Migration Quality Dashboard

CMS DE-SynPUF · Legacy → New System · Six Sigma Analysis · DuckDB + Streamlit

[← on Scorecard](#) [Six Sigma by Field](#) [Financial Impact](#) [Discrepancy Drill-Down](#) [Row Count Validation](#) [Defect Heatmap](#) [Chronic Condition Errors](#) [Year-over-Year Trend](#) [Go / No-Go Summary](#)

Go / No-Go Executive Summary

Traffic-light readiness assessment by Six Sigma threshold

Overall Migration Readiness



GO — Carrier Claims

Sigma Level: 4.563

DPMO: 1,095

Yield: 99.8904%

Meets target ($\geq 4.5\sigma$)



GO — Beneficiary Summary

Sigma Level: 4.668

DPMO: 769

Yield: 99.9231%

Meets target ($\geq 4.5\sigma$)

Field-Level Readiness — Carrier Claims

GO fields

2

CAUTION fields

4

NO-GO fields

7

[> Carrier Claims — field detail](#)

Field-Level Readiness — Beneficiary Summary

Next Steps

Error Datasets Available

Datasets showing every single field that changed after the migration were created and are available to serve the Application and Database Engineers. May help with root cause analysis (RCA).

Detailed Quality Scores Available

Detailed Quality Scores available for every field in both Beneficiary and Carrier Claims tables. Helps with setting priorities.

Analysis Positioned to Support Deep Dives

Can provide analysis to answer specific questions about the data migration that this presentation did not answer.

Analytics Improves with Time and Familiarity

Next: Interview stakeholders, gather more context about the application, data, and expectations for the migration. Then update analysis to accommodate new insights and provide custom views to answer questions that stakeholders raise.